

Anshuman Pandey

Artificial Intelligence Engineer

+91 7780527781 | a.pandeyj28@gmail.com | linkedin.com/in/anshuman-pandey | github.com/AnshumanJ28 | portfolio

PROFESSIONAL SUMMARY

Final-year B.Tech CSE (AI/ML) student at VIT Bhopal focused on building production-grade machine learning systems, not just notebook prototypes. Experience spans a tool-using LLM agent built entirely from scratch, hybrid-retrieval RAG systems with citation-grounded answers, and MLOps workflows with automated drift detection and CI/CD. Core stack is Python, PyTorch, and FastAPI, with a growing focus on reliability, testing, and deployability of ML systems. Seeking an AI/ML Engineer internship or an entry-level role.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, SQL, HTML/CSS
ML / Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, LightGBM, NumPy, Pandas, OpenCV, CNN, LSTM, ResNet, YOLOv8, timm
LLMs, Agents & RAG	ReAct Agents, Tool Calling, Framework-free Agent Design, FAISS, BM25, Reciprocal Rank Fusion, Hybrid Retrieval, Sentence-Transformers, Gemini, Groq (Llama 3.1), Hugging Face
MLOps & Deployment	FastAPI, Docker, MLflow, DVC, GitHub Actions (CI/CD), Evidently AI, Streamlit, Gradio, AWS
Concepts	Data Structures & Algorithms, A/B & Hypothesis Testing, MCTS, Reinforcement Learning, System Design Fundamentals

PROJECTS

LLM Agent From Scratch (<https://llm-agent-tez2.onrender.com>) 2026

Python, Tool Calling, Guardrails, Sandboxed Execution

- Engineered a tool-using ReAct (Reason + Act) agent entirely from scratch with no agent framework (no LangChain, LangGraph, or CrewAI), implementing a hand-rolled reasoning loop, auto-generated tool-calling schemas, and conversation/task-state memory.
- Added input/output guardrails around tool use, a Docker-sandboxed environment for the code-execution tool, and full execution tracing to debug and audit every run.
- Validated correctness with a custom evaluation harness, passing 4/4 end-to-end scenarios and 12/12 unit tests with zero regressions.

RAG Document QA – Hybrid Retrieval with Citations (<https://github.com/AnshumanJ28/rag-document-qa>) 2026

Python, FAISS, BM25, Gemini, FastAPI, Streamlit

- Designed a Retrieval-Augmented Generation system that answers questions grounded in uploaded PDFs and text documents, with page-level source citations, using PyMuPDF extraction with overlap-preserving chunking.
- Improved retrieval relevance by fusing FAISS vector search with BM25 via Reciprocal Rank Fusion and generated numbered citations mapped to source chunks using Gemini Flash.
- Built a lightweight groundedness evaluation framework using cosine similarity (no second LLM call), served through a FastAPI backend and Streamlit frontend, with a fully offline CI test suite.

TalentMatch AI – Resume/JD ATS Match Scorer (<https://talentmatch-ai-se48.vercel.app/>) 2026

Python, Groq, Sentence-Transformers, Gradio, RapidFuzz

- Developed an ATS-style resume-to-job-description scorer that outputs a 0–1 fit score with a plain-English explanation and a matched/missing/partial skills breakdown.
- Combined LLM-based resume structuring via Groq with an EasyOCR fallback for scanned PDFs, skill normalization against a canonical taxonomy using RapidFuzz, and a heuristic ranker weighting 60% cosine similarity with 40% engineered numeric features.
- Refactored an original research notebook into a production-ready pipeline, consolidating config/device/logger boilerplate and fixing a phase-ordering bug that had been silently skewing scores, then deployed it via Gradio on Hugging Face Spaces.

EXPERIENCE

Virtual Intern May 2026

ServiceNow

Remote

- Completed a virtual internship in platform administration and automation, gaining hands-on experience across 4 core areas: Flow Designer, the Automated Test Framework (ATF), Reporting, and Agentic AI concepts.

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) 2023 – Present

Vellore Institute of Technology, Bhopal

CGPA: 8.52/10

Class XII: 89.5% | Class X: 89.4%

CERTIFICATIONS

C & C++ Programming (Udemy, HackerRank) , Python & Java (HackerRank) , Cloud Computing (NPTEL) , Advanced Excel (Udemy) , AI & Agentic AI (TalentSprint) , Machine Learning (Coursera) , AWS Trainer & AWS Academy (AWS) , IoT (NPTEL) , ServiceNow

ACHIEVEMENTS

2nd Runner-up, IEEE-NSUT Ideathon – ML-based project, 80+ competing teams | **Top 10, BITS Pilani Hackathon** – national-level hackathon